

Delayed Immune-Mediated Fulminant Myocarditis Secondary to Dual Checkpoint Blockade: Mechanistic Insights and Clinical Surveillance

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ABSTRACT -- Background & Context: Immune checkpoint inhibitor (ICI) combination regimens targeting cytotoxic T-lymphocyte antigen-4 (CTLA-4) and programmed cell death protein-1 (PD-1) provide marked clinical benefit in advanced malignancies. However, immune-related adverse events (irAEs) present substantial clinical hazards, with myocarditis representing an uncommon but disproportionately fatal toxicity (fatality rates up to 50%).

Review Scope & Case Vignette: This surveillance monograph reviews institutional risk-mitigation protocols and documents an illustrative, definitive case of delayed fulminant myocarditis occurring in a 68-year-old female (Patient H.L., 62 kg) receiving Nivolumab (240 mg IV q2w) plus Ipilimumab (1 mg/kg IV q6w) for metastatic cutaneous melanoma. On Day 44 post-initiation (12 days following cycle 3), she presented with complete (third-degree) AV heart block, severe biventricular hypokinesis (LVEF 32%), marked troponin T elevation (1.84 ng/mL), and NT-proBNP 4,820 pg/mL. Urgent temporary transvenous pacing, pulse methylprednisolone (1,000 mg/day), and therapeutic plasma exchange resulted in recovery of atrioventricular conduction.

Pharmacovigilance Appraisal: Documents an identifiable, serious, life-threatening individual ICSR case embedded within comprehensive institutional surveillance data.

Keywords: Nivolumab; Ipilimumab; Delayed Immune-Mediated Fulminant Myocarditis; Third-Degree AV Block; Troponin T; ICSR; Pharmacovigilance.

1. INTRODUCTION & MECHANISTIC BASIS

Immune checkpoint blockade targeting PD-1 (nivolumab, pembrolizumab) and CTLA-4 (ipilimumab) has revolutionized modern oncology, establishing durable remissions in advanced metastatic melanoma, renal cell carcinoma, and non-small cell lung cancer [1]. These monoclonal antibodies unleash anti-tumor immunity by disinhibiting cytotoxic T-cell responses. However, non-specific T-cell activation can induce off-target autoimmune destruction against healthy host tissues, termed immune-related adverse events (irAEs). While dermatologic, endocrine, and gastrointestinal toxicities occur frequently, cardiovascular irAEs--specifically immune-mediated myocarditis--occur in approximately 0.27% to 1.14% of patients receiving dual-agent checkpoint inhibition [2].

2. EPIDEMIOLOGY & SURVEILLANCE CHALLENGES

ICI-associated myocarditis is notoriously treacherous due to its insidious onset, rapid hemodynamic destabilization, and high fatality rate (38% to 50% in combined regimens) [3]. Early clinical trials suggested that myocarditis occurs predominantly during the first 30 days of therapy. However, emerging registry data from our cardio-oncology surveillance center demonstrate that delayed presentations (>40 days post-initiation) are increasingly recognized. Because symptoms such as fatigue and exertional dyspnea mimic cancer progression, diagnosis is frequently delayed until acute cardiogenic shock or malignant arrhythmias emerge.

To illustrate the diagnostic complexity, high-resolution biomarkers, and intensive salvage therapies required, we present below a documented individual case extracted from our prospective institutional registry.

3. CLINICAL CASE VIGNETTE: PATIENT H.L.

Patient History & Baseline: Patient H.L. is a 68-year-old Caucasian female (body weight 62 kg) who was diagnosed with unresectable, BRAF wild-type stage IV metastatic cutaneous melanoma with pulmonary and retroperitoneal nodal metastases. Her medical history was notable for mild osteoporosis and well-controlled hypercholesterolemia. Baseline electrocardiogram (ECG) demonstrated normal sinus rhythm with normal intervals (PR 156 ms, QRS 84 ms, QTc 418 ms), and baseline transthoracic echocardiogram revealed normal left ventricular ejection fraction (LVEF 62%) without regional wall motion abnormalities. Serum troponin T and NT-proBNP were normal.

Immunotherapy Exposure: First-line dual immunotherapy was commenced with Nivolumab (Opdivo) 240 mg IV every 2 weeks combined with Ipilimumab (Yervoy) 1 mg/kg IV every 6 weeks. She tolerated cycles 1 and 2 without immune-mediated toxicity.

Acute Presentation (Day 44): On Day 44 post-initiation (exactly 12 days following cycle 3), Patient H.L. presented to the emergency department with a 3-day history of progressive exertional dyspnea, orthopnea, presyncope, and profound generalized weakness. Vital signs: blood pressure 88/52 mmHg, heart rate 38 bpm (severe bradycardia), respiratory rate 22 breaths/min, and oxygen saturation 92% on room air. Jugular venous distension was elevated to 10 cm H2O, and bilateral basilar crackles were audible. There was no chest pain, fever, or rash.

Diagnostic Cardiodiagnostics: Emergency 12-lead ECG revealed **Complete (Third-Degree) AV Heart Block with a junctional escape rhythm** at 38 bpm (QRS 132 ms). Bedside echocardiogram demonstrated acute cardiac decompensation with an acute LVEF decline to 32% and diffuse biventricular hypokinesis. Serum cardiac biomarkers were severely elevated: high-sensitivity cardiac Troponin T was 1.84 ng/mL (ref <0.014 ng/mL, >130x ULN), and N-terminal pro-b-type natriuretic peptide (NT-proBNP) was 4,820 pg/mL (ref <125 pg/mL). Creatine kinase-MB was 68 ng/mL (ref <5 ng/mL).

Table 1. Longitudinal Cardiac Biomarkers & Conduction Parameters in Patient H.L.

Parameter	Reference	Baseline	Day 44 (Peak)	Day 63 (Disch.)
Troponin T (ng/mL)	<0.014	0.008	1.84 (>130x)	0.042
NT-proBNP (pg/mL)	<125	84	4,820	340
CK-MB (ng/mL)	<5.0	1.8	68.0	3.2
LVEF (%)	>55%	62%	32% (Severe)	48% (Recovered)
Rhythm / Conduction	Sinus	Sinus (PR 156)	Complete AV Block	Sinus (PR 182)

4. INVASIVE MONITORING, MANAGEMENT & CLINICAL COURSE

Patient H.L. was urgently transferred to the Cardiac Intensive Care Unit (CICU). Immunotherapy was immediately suspended permanently (positive dechallenge). Due to hemodynamically compromising complete heart block and hypotension refractory to atropine and dopamine infusion, an urgent temporary transvenous pacemaker was placed via the right internal jugular vein, stabilizing heart rate at 80 bpm. Coronary angiography was immediately performed and demonstrated widely patent coronary arteries without obstructive atheroma or plaque rupture, ruling out acute coronary syndrome.

High-dose pulse intravenous methylprednisolone (1,000 mg IV daily) was initiated within 2 hours of presentation. On hospital day 2, a transvenous right ventricular endomyocardial biopsy was performed. Histopathology demonstrated **dense interstitial infiltration of CD4+ and CD8+ T-lymphocytes accompanied by multifocal cardiomyocyte necrosis** and interstitial edema (Figure 1), confirming high-grade immune-mediated myocarditis. Because Troponin T remained elevated (>1.5 ng/mL) at 48 hours, therapeutic plasma exchange (TPE; 5 sessions on alternate days) was instituted as secondary immunosuppression [4].

5. CLINICAL RESOLUTION & OUTCOME

Following completion of pulse corticosteroids and TPE, troponin T steadily declined to 0.042 ng/mL by day 18 (Table 1). Normal sinus rhythm with 1:1 AV conduction restored on hospital day 8, permitting successful removal of the temporary transvenous pacemaker without need for permanent pacemaker implantation. Repeat echocardiogram prior to discharge demonstrated substantial recovery of left ventricular systolic function (LVEF 48%). She was transitioned to oral prednisone (1 mg/kg/day) with a planned slow 8-week taper. At 90-day follow-up, she remained free of cardiac symptoms with stable oncologic disease. Immunotherapy was permanently discontinued.

6. PHARMACOVIGILANCE & EVIDENCE RETRIEVAL APPRAISAL

This article represents a classic challenging literature screening benchmark: an identifiable individual safety report (Patient H.L., 68F, nivolumab + ipilimumab, delayed fulminant myocarditis with complete heart block) embedded within broader institutional review narrative. Automated screening engines must successfully identify the patient case despite introductory epidemiological context. Under ICH E2D and FDA 21 CFR 314.80, this event is classified as **Serious**, meeting both **Life-Threatening** and **Inpatient Hospitalization** criteria, requiring mandatory expedited ICSR reporting. Evidence is distributed across pages (Case presentation on Page 1; Invasive management, biopsy, and clinical resolution on Page 2).

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